



EXPERIMENT NO 4

INDUCTION MACHINE AS GENERATOR AND MOTOR

AIM

- To operate the given 3 phase induction machine as a) induction motor and b) induction generator
- To conduct load test in both generating and motor modes
- To plot the performance characteristics
- To plot W vs slip and hence determine the hysteresis power.

APPARATUS

Sl No.	Name of the apparatus	Type	Range	Quantity
1.	Voltmeter			
2.	Ammeter			
3.	Wattmeter			
4.	Rheostat			
5.	Stopwatch			
6.	Tachometer			

PRINCIPLE

An induction generator is asynchronous in nature because of which it is commonly used as windmill generator since a windmill runs at non-fixed speed. These are used in remote areas to supplement power received from weak transmission links.

If the induction machine is driven at a speed greater than synchronous speed by a prime mover, the direction of induced torque reverses and it acts as an induction generator. The rotating magnetic field is set up by the magnetizing current drawn from the mains. Based on the way with which the generator gets the required lagging reactive-power, they are classified into i) Line excited induction generator and ii) Self-excited induction generator.

Line excited induction generator: Induction machine connected to supply mains and driven at super-synchronous speed by its prime mover is called a line excited induction generator. The generator draws the required lagging reactive power from the mains.

Self excited induction generator: In this generators, the necessary lagging reactive power for its excitation is obtained by a capacitor bank connected across the generator terminals. If the rotor of the machine is driven by its prime-mover, the presence of residual flux (present in rotor core) causes a small emf to get induced in the stator windings at a frequency proportional to the rotor speed. This voltage impresses over the 3-phase capacitor gives rise to leading current drawn by the capacitor which is equivalent to lagging current supplied to the generator. The flux set up by this current assists the initial residual flux. Hence the net flux will increase causing the voltage to build up farther. The